

Tutorial of potential of mean force by MolAICal

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1. Introduction

The potential of mean force (PMF) can be used to calculate the free energy landscape with principal components. The PMF along the coordinate is computed from the average distribution function (see below equation).

$$\Delta G = -k_B T \ln \rho(x, y)$$

Where T and k_B is the temperature and Boltzmann constant, respectively. The x and y represent two principal components. In this tutorial, the molecular dynamics (MD) simulated results of glucagon receptor (GCGR) are selected for this example (Front Chem. 2019 Dec 17;7:851.)¹. Here, MolAICal (<https://molaical.github.io>) is employed for this tutorial.

2. Materials

2.1. Software requirement

1) MolAICal: <https://molaical.github.io>

2.2. Example files

1) All the necessary tutorial files are downloaded from:
<https://github.com/MolAICal/tutorials/tree/master/007-PMF>

3. Procedure

This tutorial can be run on Linux and Windows systems. In the Linux system, it may happen errors as follows:

Exception in thread "main" java.lang.UnsatisfiedLinkError: /home/feng/MolAICal-linux64-v1.1/libdislin_d.so.11: libXm.so.4: cannot open shared object file: No such file or directory

Solution: It is lack of openmotif. Users can install it by root account:

```
sudo apt-get update -y  
sudo apt-get install -y libmotif-common  
sudo apt-get install libmotif-dev
```

3.1. Plot energy contour by MolAICal

```
#> cd 007-PMF
```

Open “rmsd-dis.dat”, the 1st column is RMSD values, 2nd column is the distance. You can replace these data with your appointed principal components. Then, run the command:

```
#> molaical.exe -pmf -i rmsd-dis.dat
```

The plotted results show in Figure 1

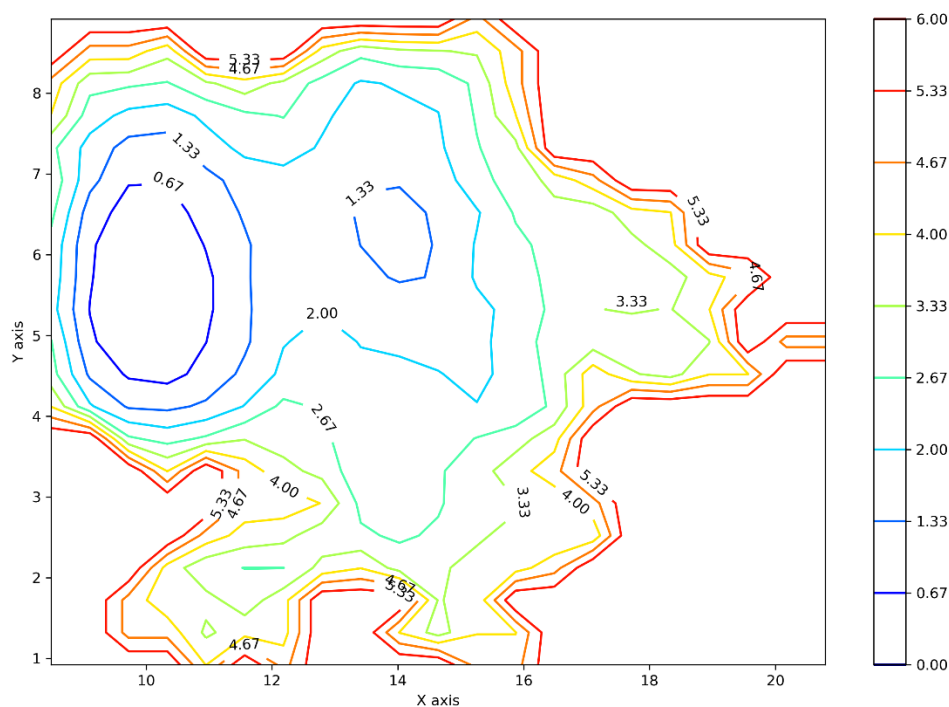


Figure 1. PMF contour (Unit of free energy landscape: kcal/mol)

Running the command as below. It plots the figure with another shape (see Figure 2).

```
#> molaical.exe -pmf -i rmsd-dis.dat -g 20 -l 10 -m conshd -b none -x "RMSD (Å)" -y "Distance (Å)"
```

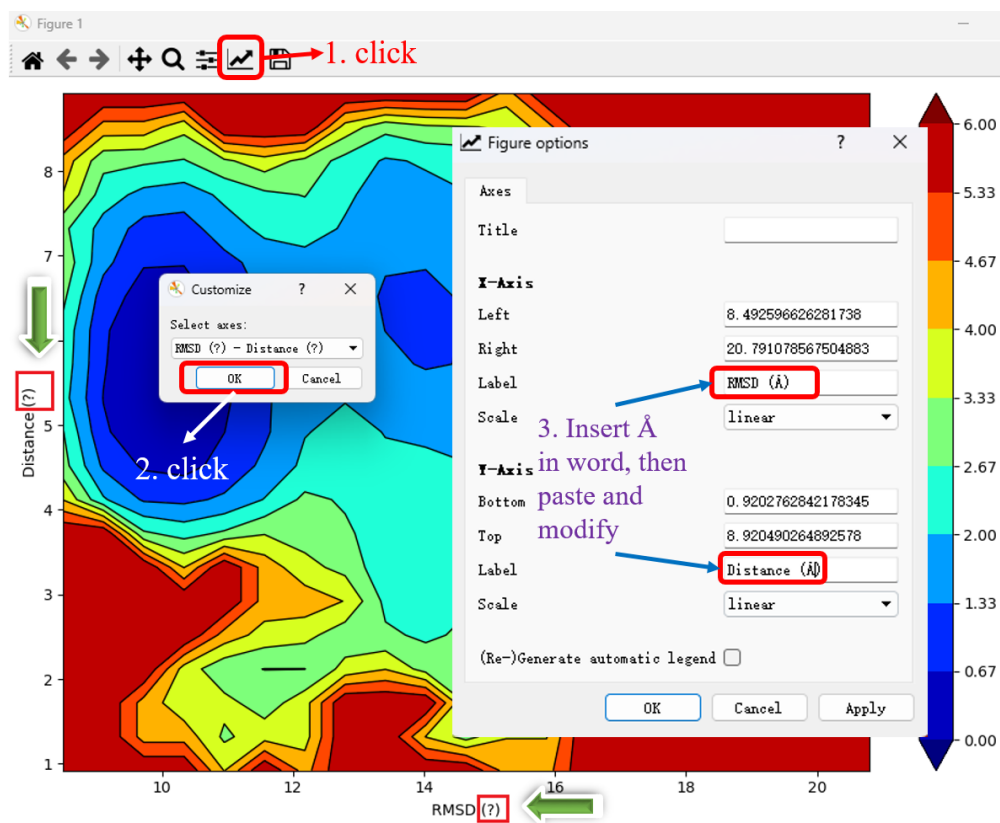


Figure 2. PMF contour (Unit of free energy landscape: kcal/mol)

Note: As shown in Figure 2, executing the above command will pop up an image display window. Sometimes the command line may fail to recognize the character 'Å' and display it as a garbled character '?'. In such cases, users can correct the garbled character by operating the options in the image display window (refer to the specific operation steps in Figure 2).

If users want to add numeric labels to contour lines in 'conshd' mode, they can use the following command:

```
#> molaical.exe -pmf -i rmsd-dis.dat -g 20 -l 10 -m conshd -b float -x "RMSD (Å)" -y "Distance (Å)"
```

Executing the above command will generate Figure 3:

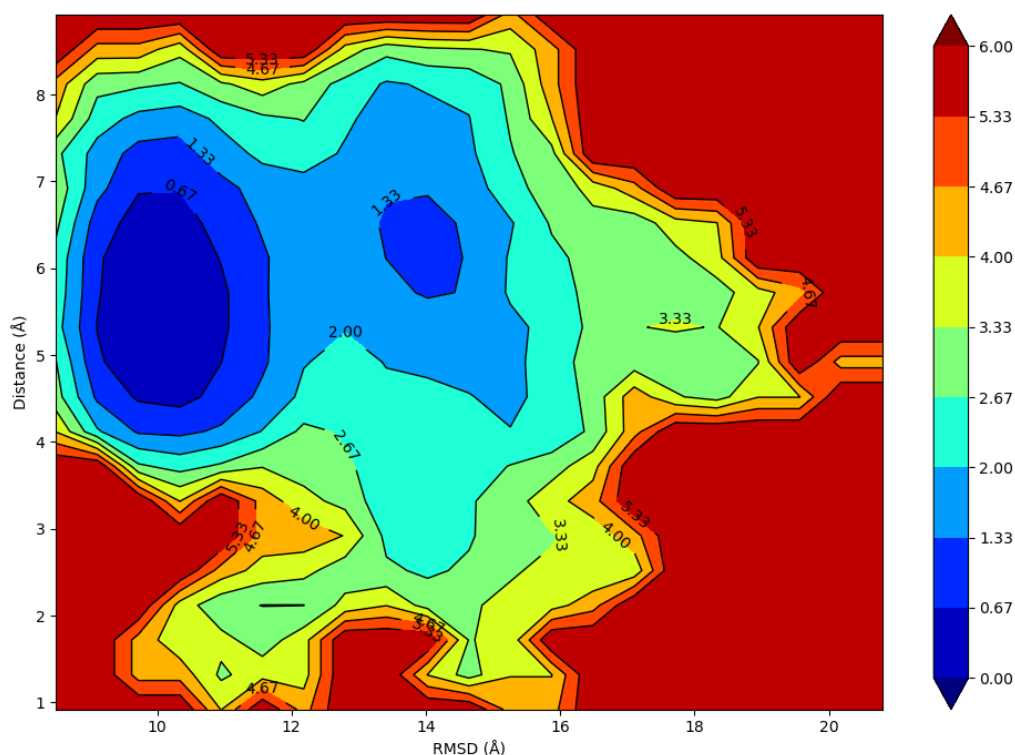


Figure 3. PMF contour with labeled isopleths, showing the free energy landscape (free energy unit: kcal/mol).

The command parameters of MolaICal for PMF calculation are as below:

- pmf:** means PMF calculation function.
- i:** the path of the input data file.
- g:** number of grids for density calculation in a square.
- l:** number of color levels.
- m:** choose contours. Currently, it contains “conshd” and “contur”.
- b:** is LABELS. It determines which label types will be plotted on an axis. It contains “none” and “float”. The default value is “float”.
 - ✧ “none” means no numeric labels are marked on the contour lines.
 - ✧ “float” represents that numeric labels are marked on the contour lines.

-x: is x-axis label.
-y: is y-axis label.
-t: is temperature.
-f: is the output figure file, the default figure name is "pmfFig.png".

For more detailed commands about PMF, please check the MolAICal manual.

3.2. Custom Plotting

3.2.1. Modify the parameters or codes for plotting

When executing the above command line, a file named “**pmf_data_plot.dat**” will be generated in the current folder by default. This file is used for custom plotting. This tutorial provides a Matplotlib plotting script named “**matplotlib_pmf.py**”. Users can achieve customized plotting by modifying certain variables or code. For example, the following code can be adjusted for customization:

```
# From matplotlib_pmf.py
# Define plotting mode
pmf_draw_mode = 'contur'
# Define the number of contour lines
pmf_color_levels = 10
# Specify whether to label numbers on contour lines
pmf_fig_label = 'float'
# Define x-axis label
pmf_x_label = 'x-axis'
# Define y-axis label
pmf_y_label = 'y-axis'
# Define output filename
pmf_fig_name = 'pmf_figure.png'
# Define the path to the output data file pmf_data_plot.dat
data = np.loadtxt('pmf_data_plot.dat')
```

Note: Users can further modify the matplotlib_pmf.py script for more refined plotting.

3.2.2. Plot the PMF figure in MolAICal

To plot the PMF figure in MolAICal, go to the virtual environment:

✧ In the Windows system, open DOS, input the below command, and press “Enter” on the keyboard:

```
#> D:\MolAICal-win64\mtools\py\Scripts\activate.bat
```

✧ In the Linux system, open Linux Terminal, input the below command, and press “Enter” on the keyboard:

```
#> source /home/feng/tutorial/MolAICal-linux64/mtools/py/bin/activate
```

Note: Please replace “D:\MolAICal-win64” or “/home/feng/tutorial/MolAICal-linux64” with your **real path** of MolAICal.

If you go into the virtual environment, users will see a graph similar to Figure 4:

```
(py) feng@feng-System-Product-Name:~/tutorial$ ls *dat
deal_rec_list.dat  lig_list.dat  lig_list_mol2.dat  rec_list.dat
(py) feng@feng-System-Product-Name:~/tutorial$
```

Figure 4

Go to the folder “007-PMF”:

```
#> cd 007-PMF
```

Then, input the below command to draw and save the PMF plotting figure (see Figure 5):

```
#> python matplotlib_pmf.py
```

Note: Users can **modify** the file “matplotlib_pmf.py” for specific graphic requirements if familiar with Matplotlib (<https://matplotlib.org>).

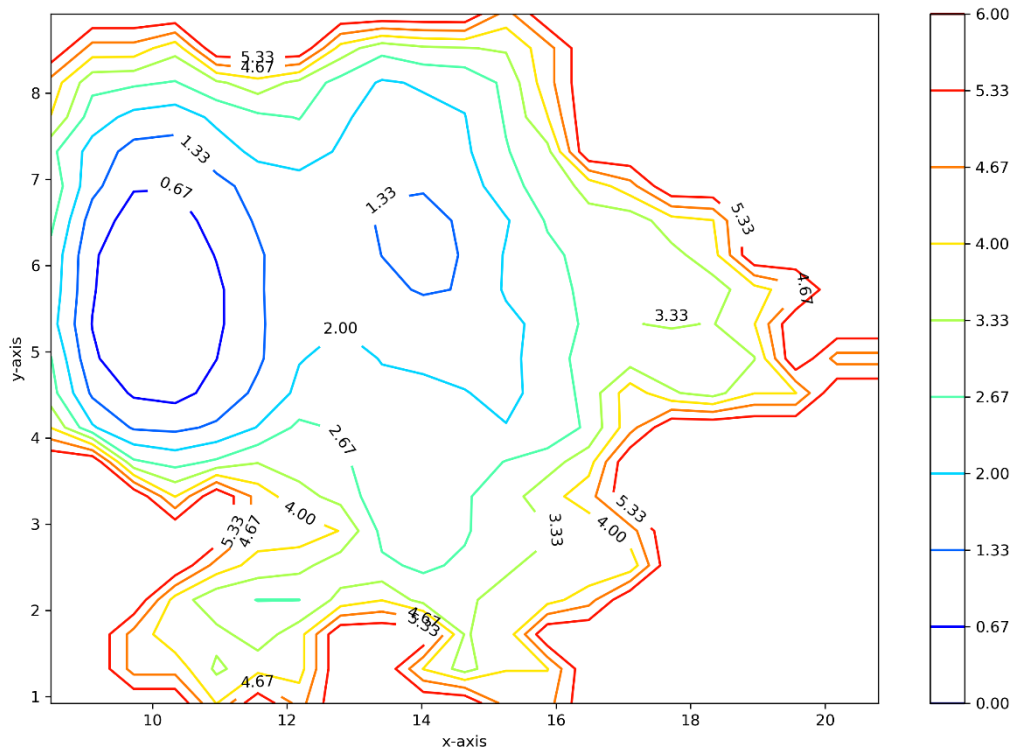


Figure 5

Appendix I

The part just produces a beautiful figure by OriginLab software. If you do not care about it, you can skip this part. The demo version of OriginLab can be downloaded from <https://www.originlab.com>.

The default generated data file “pmf_data_plot.dat” can be used to reproduce the contour of free energy.

- 1) Import “pmf_data_plot.dat” (see Figure a1)

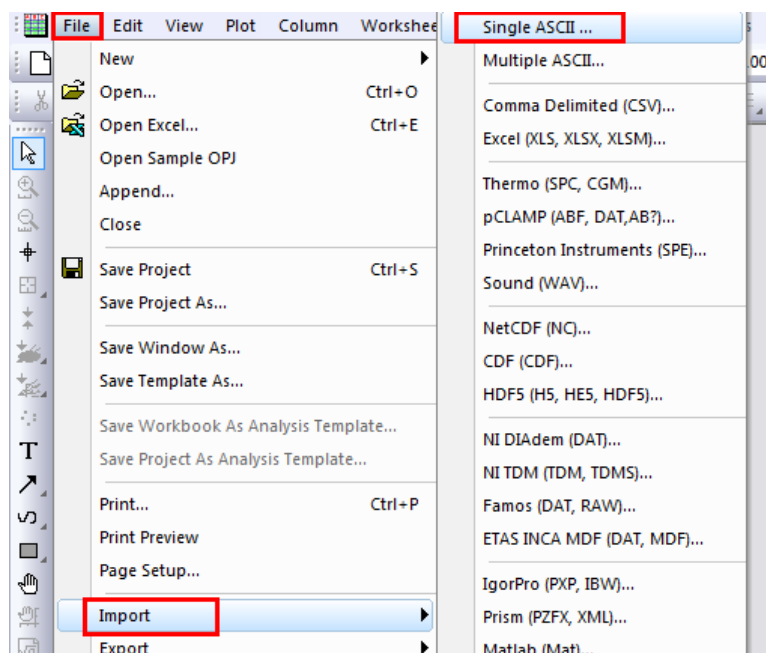


Figure a1. Import data

- 2) Double click selected column C(Y) and change C(Y) to Z (see Figure a2)

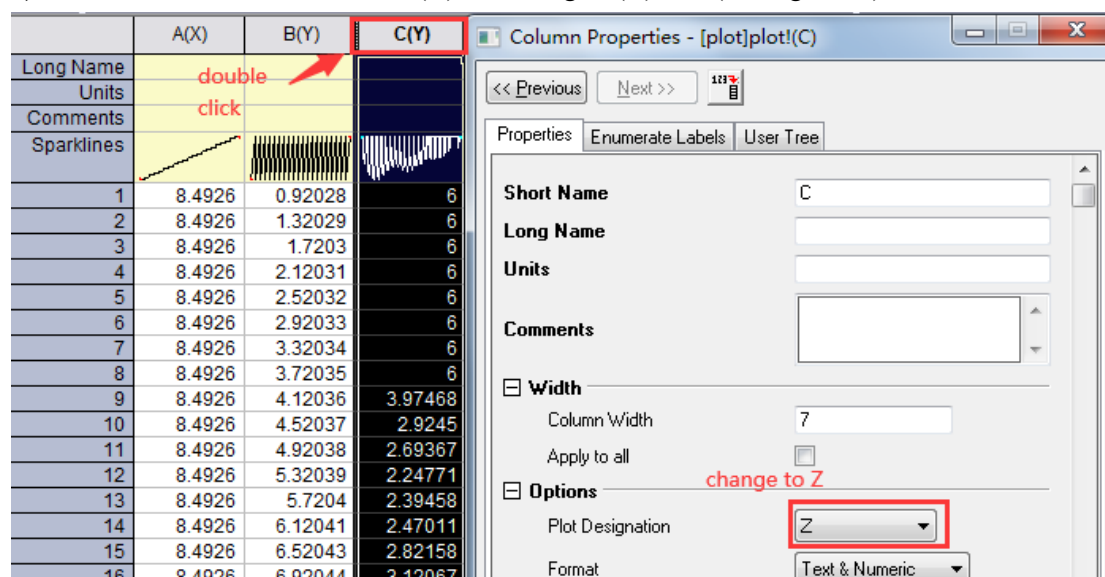


Figure a2. Set parameters for plotting

- 3) Select all data columns and plot the contour (see Figure a3)

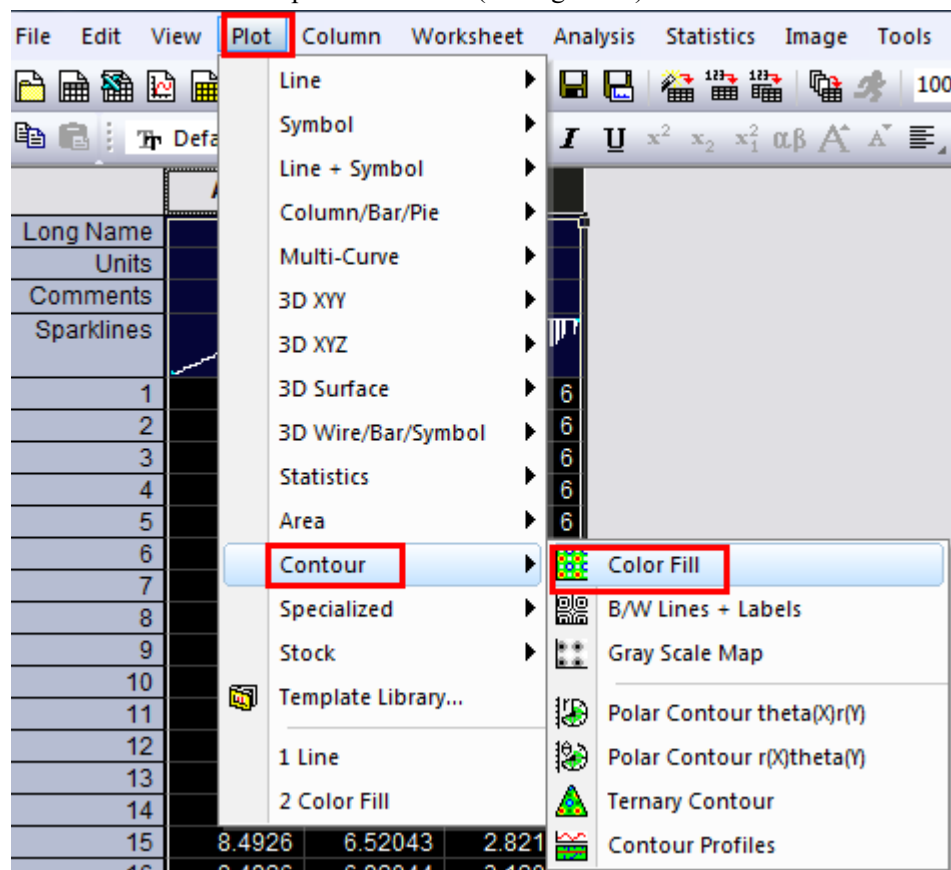


Figure a3. Plot contour

- 4) It may show “Speed Mode is On”. You can double click the contour, click “Layer1”, choose “Size/Speed”, and cancel the term in the red box in Figure a4. If you do not want to cancel “Speed Mode”, you can skip this step.

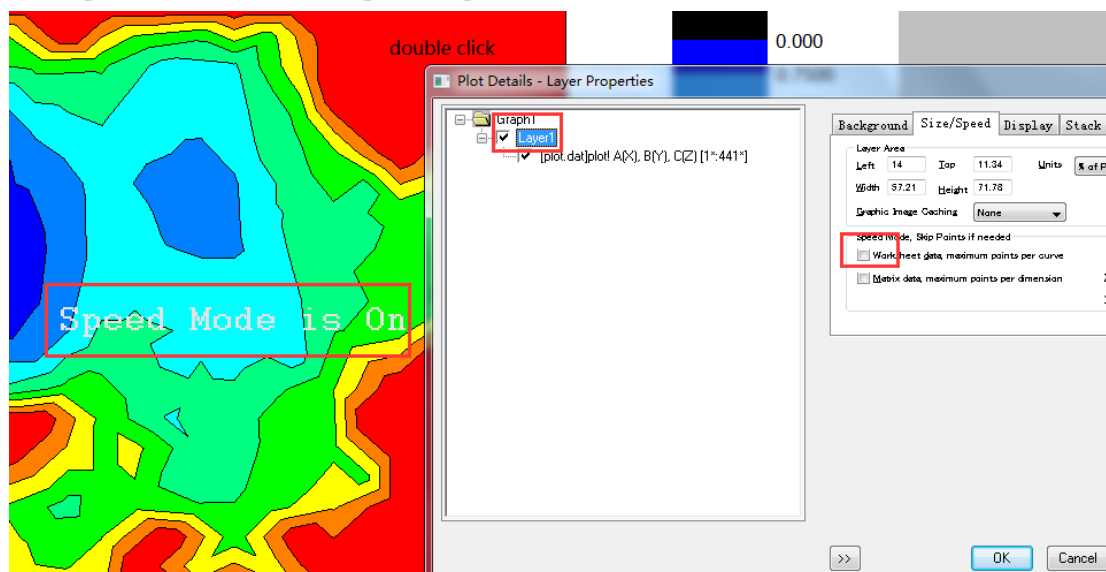


Figure a4. Cancel “Speed Mode”

- 5) If you want to show values on the contour lines, users can double click mouse on the contour

and select the terms as shown in Figure a5.

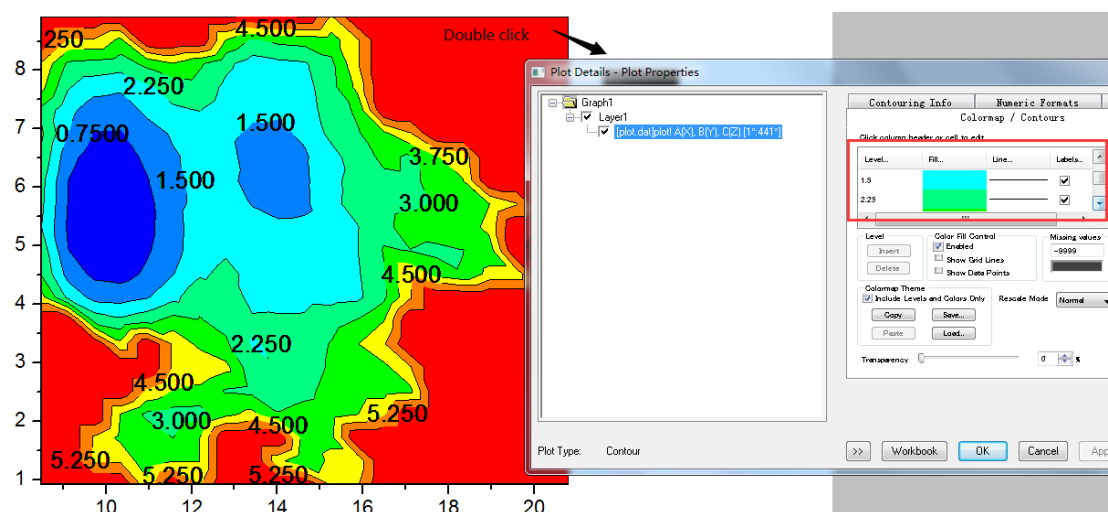


Figure a5. Plotting contour (Unit of free energy landscape: kcal/mol)

References

- 1 Bai, Q. *et al.* Conformation Transition of Intracellular Part of Glucagon Receptor in Complex With Agonist Glucagon by Conventional and Accelerated Molecular Dynamics Simulations. *Front Chem* **7**, 851 (2019).